

Lecture 12: Data Visualization Case Study

Heather Mattie, PhD

October 9, 2025

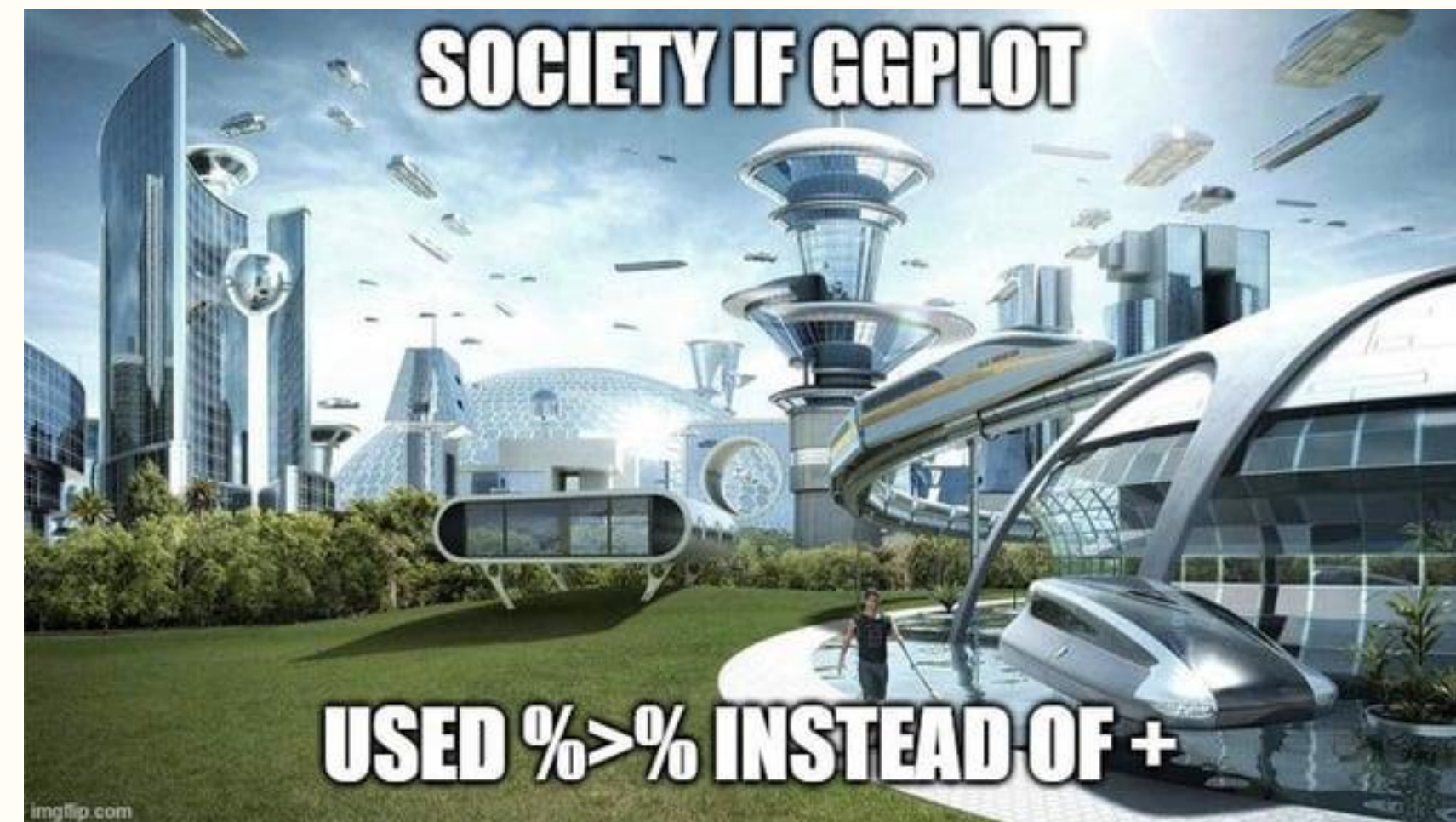
Recipe of the Day!

Epic Charcuterie Board



Announcements

- Homework #2 due Friday, October 10 by 11:59pm
- Lab will be held this week



Coding question of the day

Using the `gapminder` dataset, create a line graph of `life_expectancy` over time for the country Haiti. Make sure to include informative axis labels and a title.

There is a substantial decrease in life expectancy one year - look up the reason for this if you don't know it already. Add text to your plot with `geom_text` indicating the event that happened that year that had such a drastic impact on life expectancy.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data(gapminder)
```

Data visualization case study

Data Visualization Case Study Roadmap

1. Gapminder case study
2. Time series plots
3. Transformations

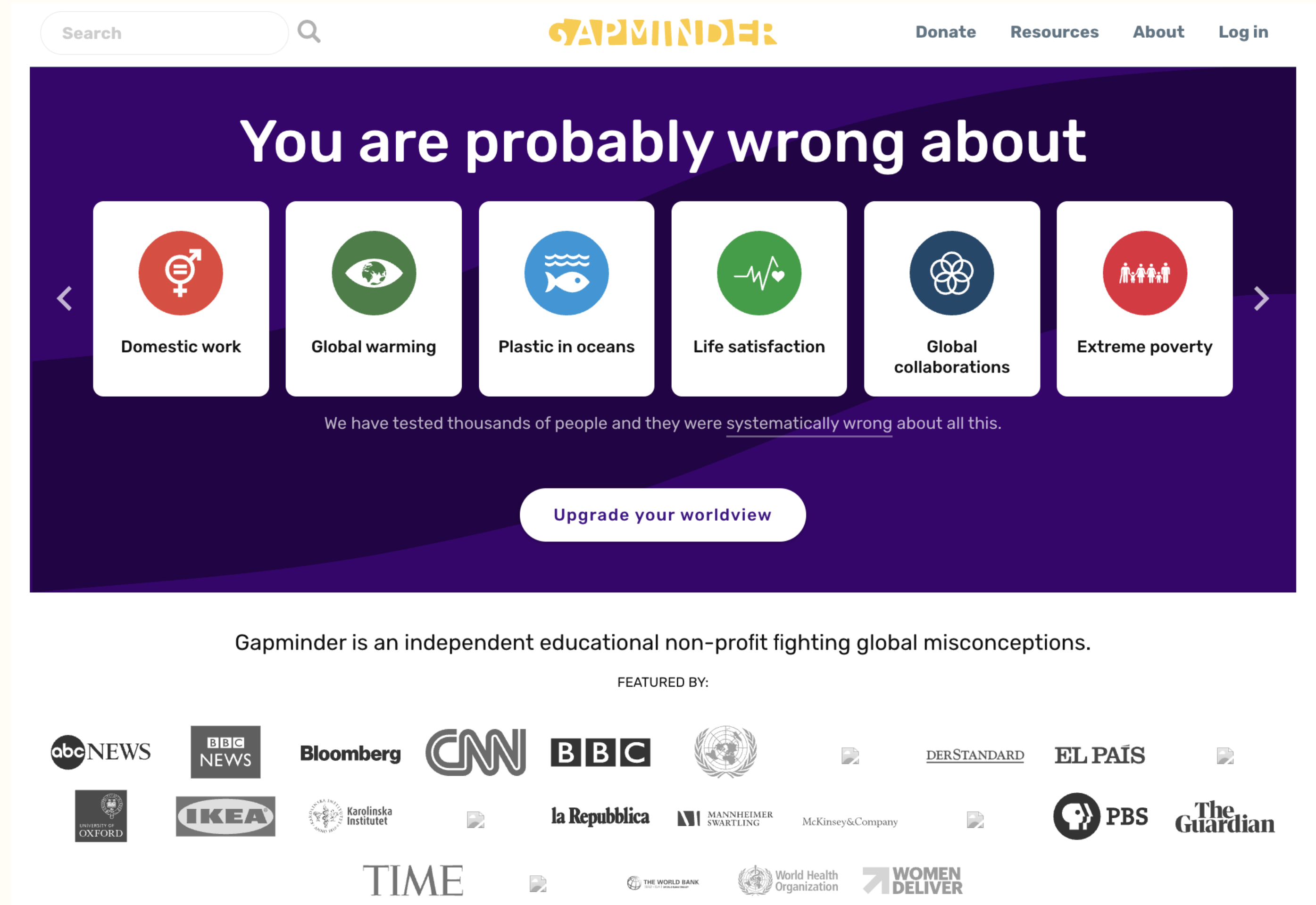
Data visualization case study

Gapminder case study

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Gapminder

- Organization co-founded by Hans Rosling
- Dedicated to educating the public by using data to dispel common myths about the so-called developing world



Hans Rosling's pre-test

For each of the five pairs of countries below, which country do you think had the **highest child mortality in 2015**?

Sri Lanka	Turkey
Poland	South Korea
Malaysia	Russia
Pakistan	Vietnam
Thailand	South Africa

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Students: 1.8 ± 0.4



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Chimpanzees: 2.5



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Sri Lanka	Turkey 
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Malaysia	Russia 
Pakistan 	Vietnam
Thailand	South Africa 

Students: 1.8 ± 0.4

Chimpanzees: 2.5

Faculty: 2.4 ± 0.4



Not ignorance – preconceived ideas

1. Is it a fair characterization of today's world to say it is divided into Western “rich” nations and the “developing world” in Africa, Asia, and Latin America?
2. Has income inequality across countries worsened the last 40 years?

gapminder dataset

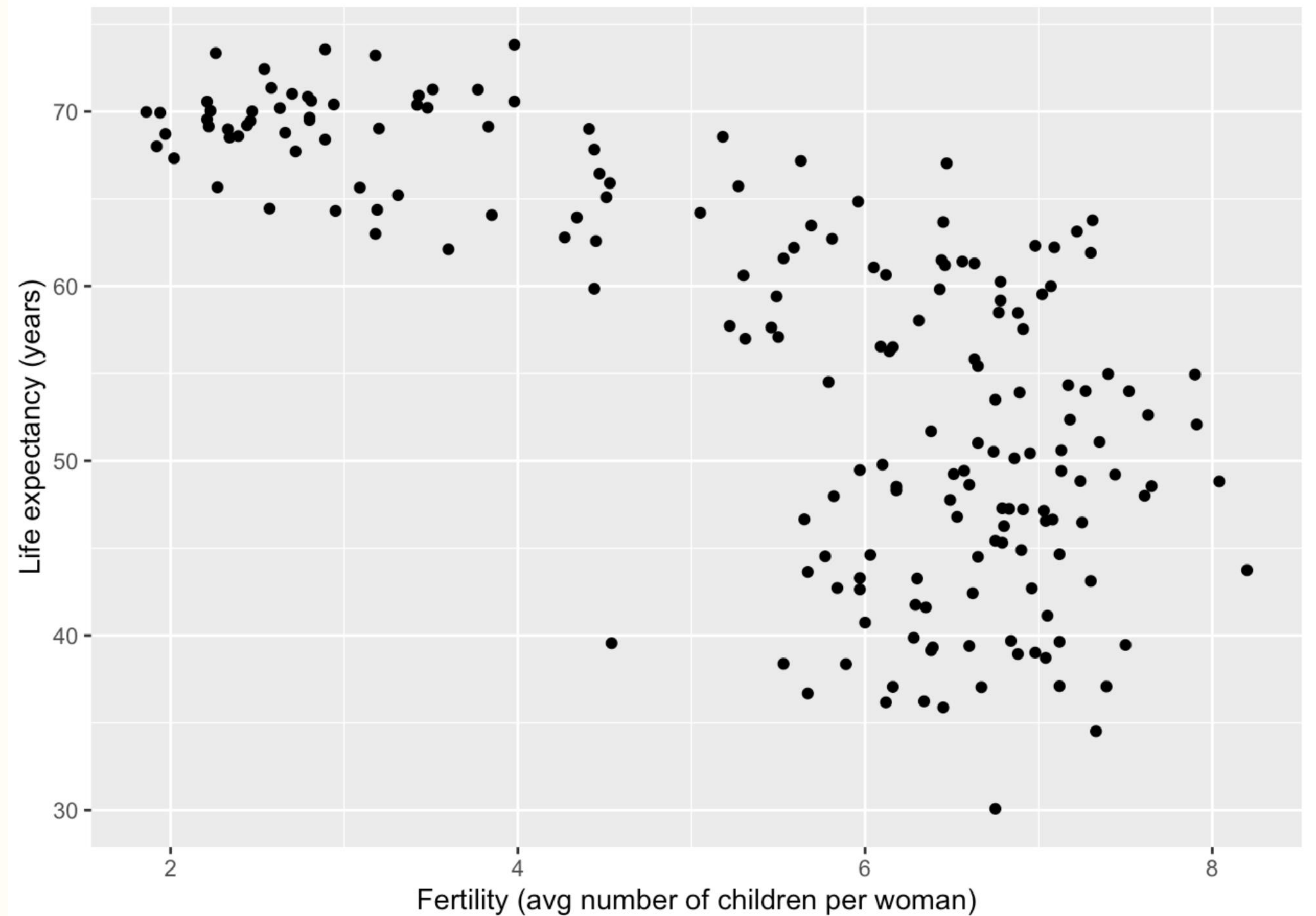
```
library(dslabs)
data(gapminder)
head(gapminder)
```

##	country	year	infant_mortality	life_expectancy	fertility
## 1	Albania	1960	115.40	62.87	6.19
## 2	Algeria	1960	148.20	47.50	7.65
## 3	Angola	1960	208.00	35.98	7.32
## 4	Antigua and Barbuda	1960	NA	62.97	4.43
## 5	Argentina	1960	59.87	65.39	3.11
## 6	Armenia	1960	NA	66.86	4.55
##	population	gdp	continent	region	
## 1	1636054	NA	Europe	Southern Europe	
## 2	11124892	13828152297	Africa	Northern Africa	
## 3	5270844	NA	Africa	Middle Africa	
## 4	54681	NA	Americas	Caribbean	
## 5	20619075	108322326649	Americas	South America	
## 6	1867396	NA	Asia	Western Asia	



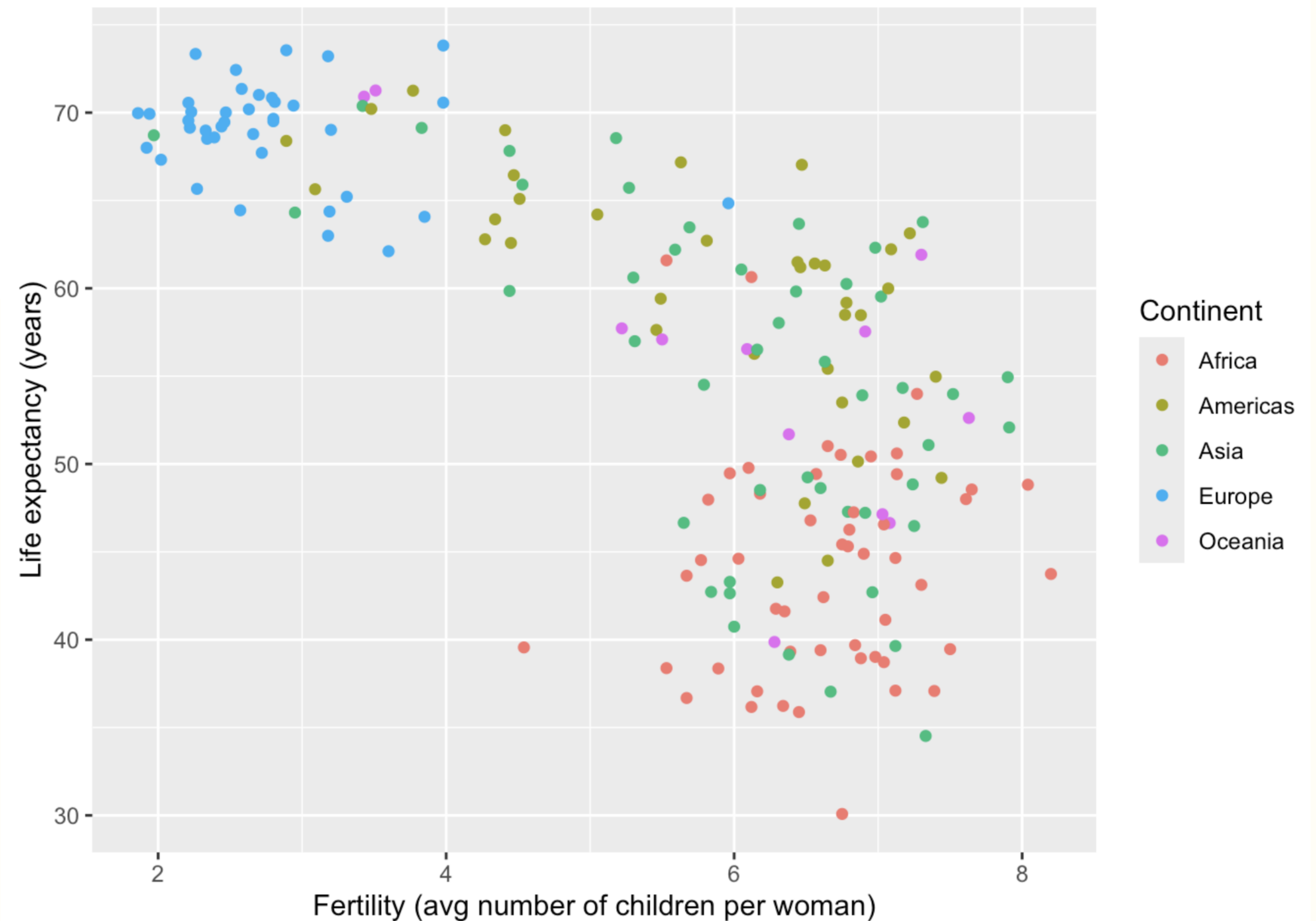
Life expectancy and fertility

```
gapminder %>% filter(year == 1962) %>%  
  ggplot(aes(fertility, life_expectancy)) +  
  geom_point() +  
  ylab("Life expectancy (years)") +  
  xlab("Fertility (avg number of children per woman)")
```



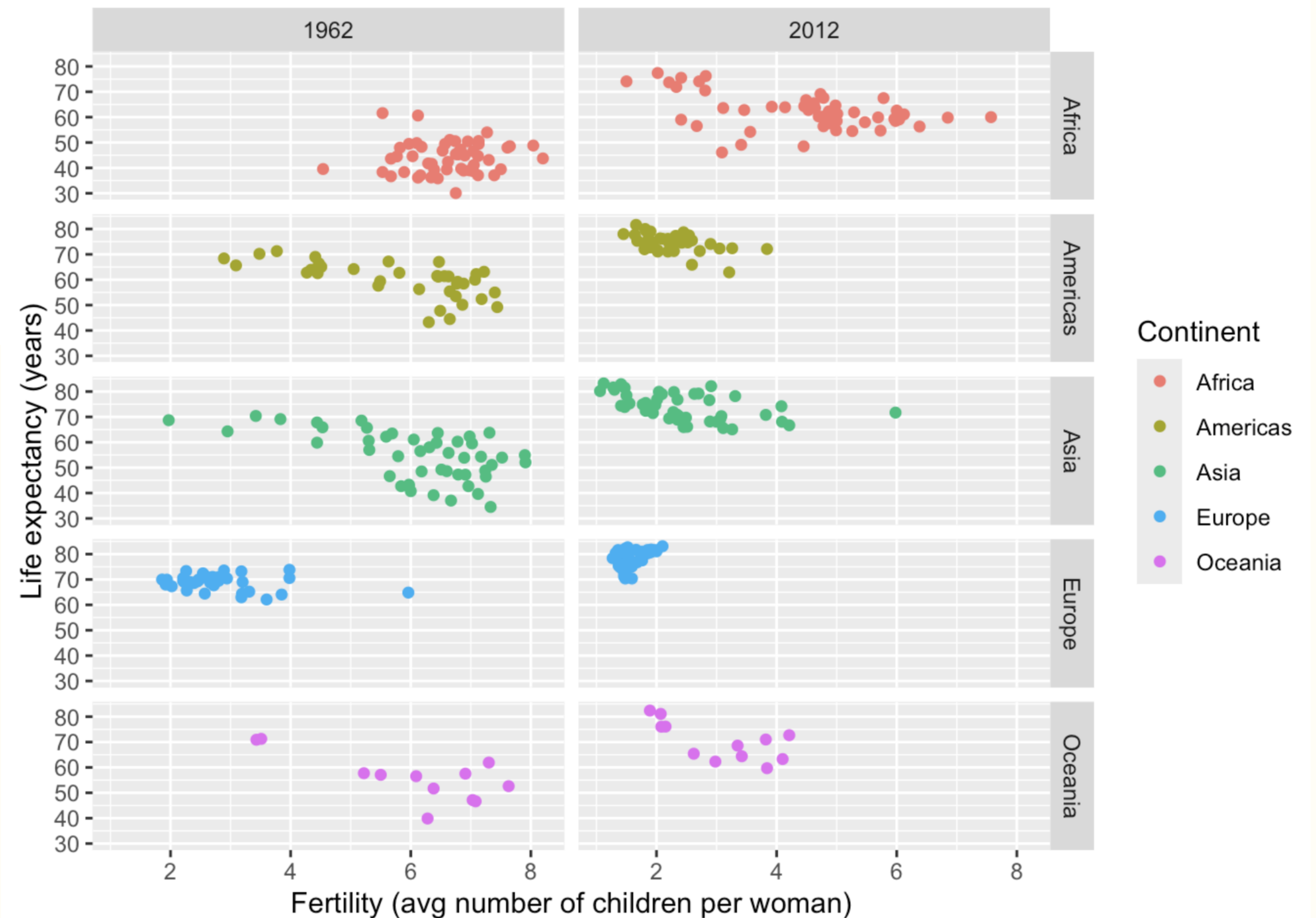
Life expectancy and fertility

```
gapminder %>% filter(year == 1962) %>%  
  ggplot(aes(fertility, life_expectancy, color = continent)) +  
  geom_point() +  
  xlab("Fertility (avg number of children per woman)") +  
  ylab("Life expectancy (years)") +  
  scale_color_discrete(name = "Continent")
```



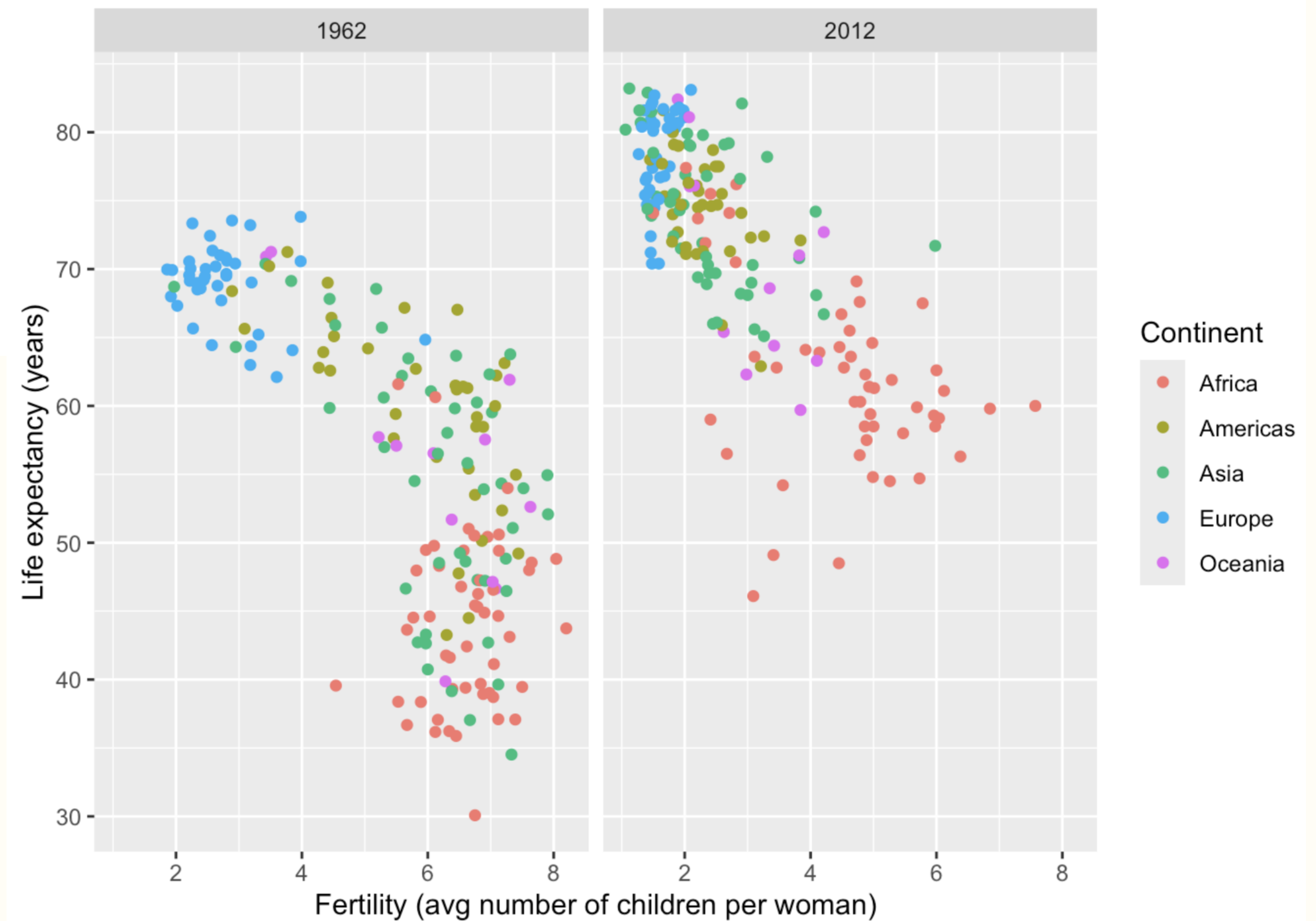
Life expectancy and fertility

```
gapminder %>% filter(year %in% c(1962, 2012)) %>%  
  ggplot(aes(fertility, life_expectancy, color = continent)) +  
  geom_point() +  
  facet_grid(continent~year) +  
  xlab("Fertility (avg number of children per woman)") +  
  ylab("Life expectancy (years)") +  
  scale_color_discrete(name = "Continent")
```



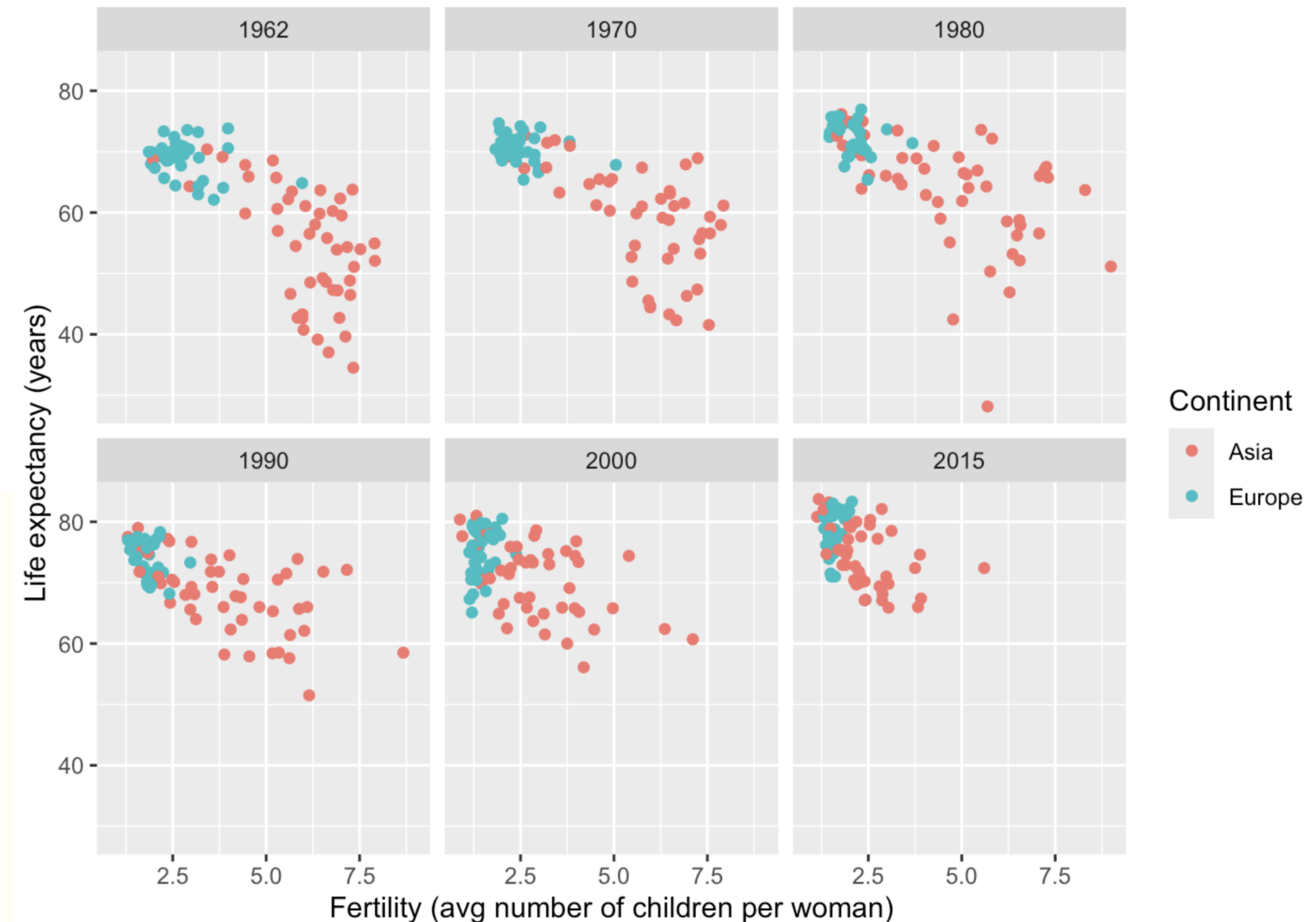
Life expectancy and fertility

```
gapminder %>% filter(year %in% c(1962, 2012)) %>%  
  ggplot(aes(fertility, life_expectancy, color = continent)) +  
  geom_point() +  
  facet_grid(. ~ year) +  
  xlab("Fertility (avg number of children per woman)") +  
  ylab("Life expectancy (years)") +  
  scale_color_discrete(name = "Continent")
```



Life expectancy and fertility

```
years <- c(1962, 1970, 1980, 1990, 2000, 2015)
continents <- c("Europe", "Asia")
gapminder %>%
  filter(year %in% years & continent %in% continents) %>%
  ggplot(aes(fertility, life_expectancy, color = continent)) +
  geom_point() +
  facet_wrap(. ~ year) +
  xlab("Fertility (avg number of children per woman)") +
  ylab("Life expectancy (years)") +
  scale_color_discrete(name = "Continent")
```



Summary

- Data visualization can be used to test preconceived ideas
- Differences across groups can become more apparent by using color
- Differences over time can be visualized with faceting

Data visualization case study

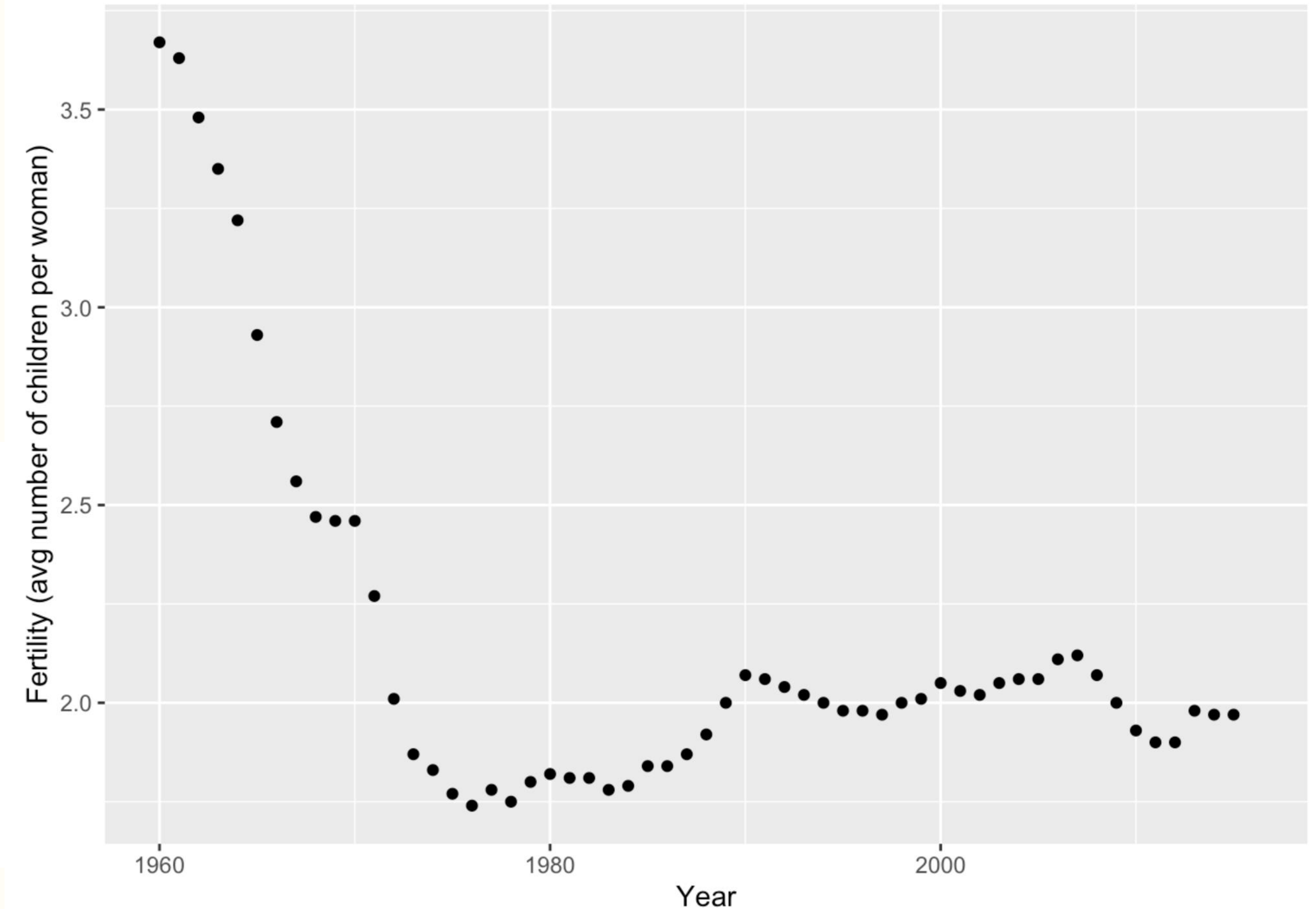
Time series plots

Heather Mattie, PhD

Time series plots

Time series plots have time on the x-axis and an outcome or measurement of interest on the y-axis

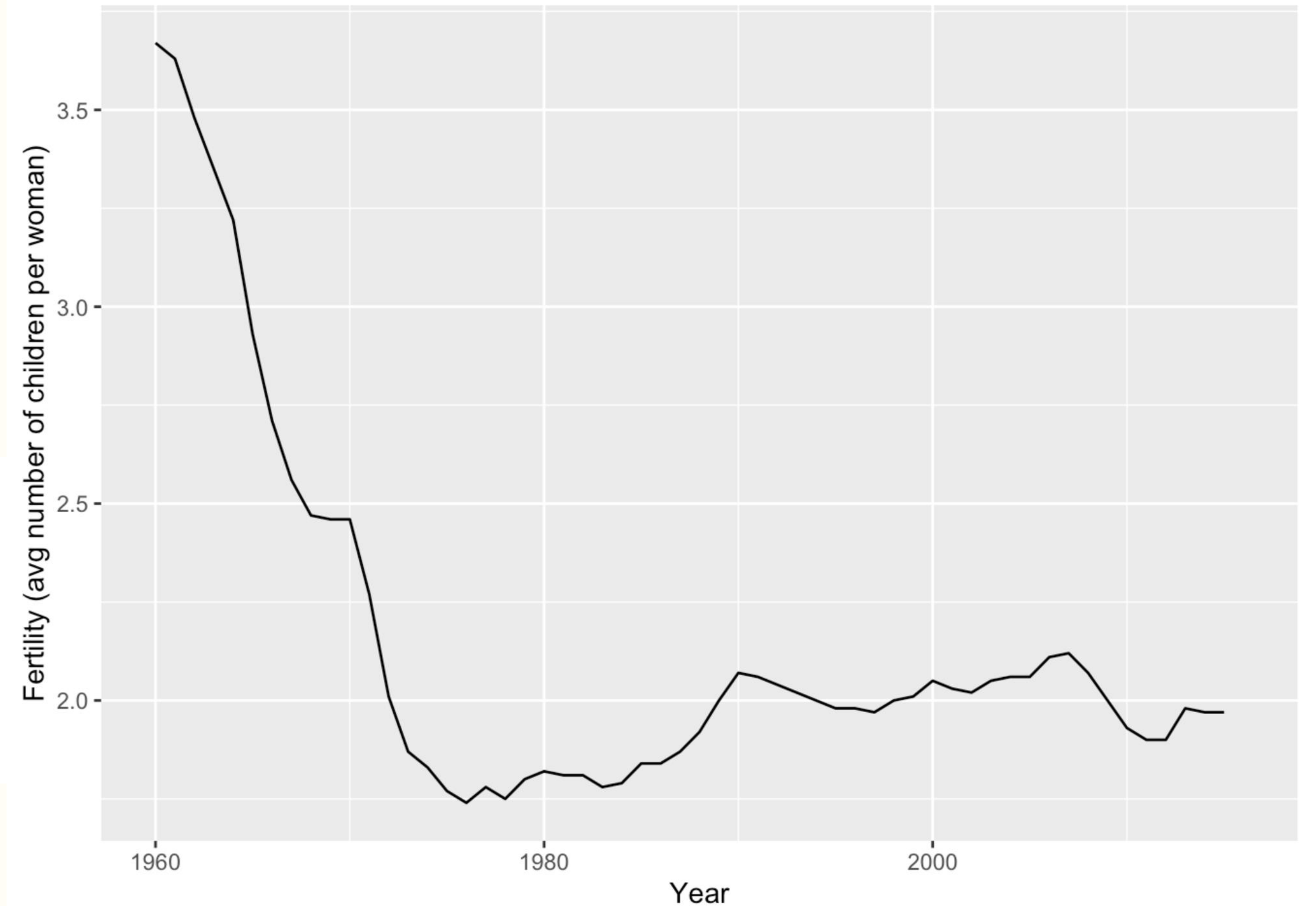
```
library(dslabs)
library(tidyverse)
gapminder %>% filter(country == "United States") %>%
  ggplot(aes(year, fertility)) +
  geom_point() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```



Time series plots

Time series plots have time on the x-axis and an outcome or measurement of interest on the y-axis

```
gapminder %>% filter(country == "United States") %>%  
  ggplot(aes(year, fertility)) +  
  geom_line() +  
  xlab("Year") +  
  ylab("Fertility (avg number of children per woman)")
```

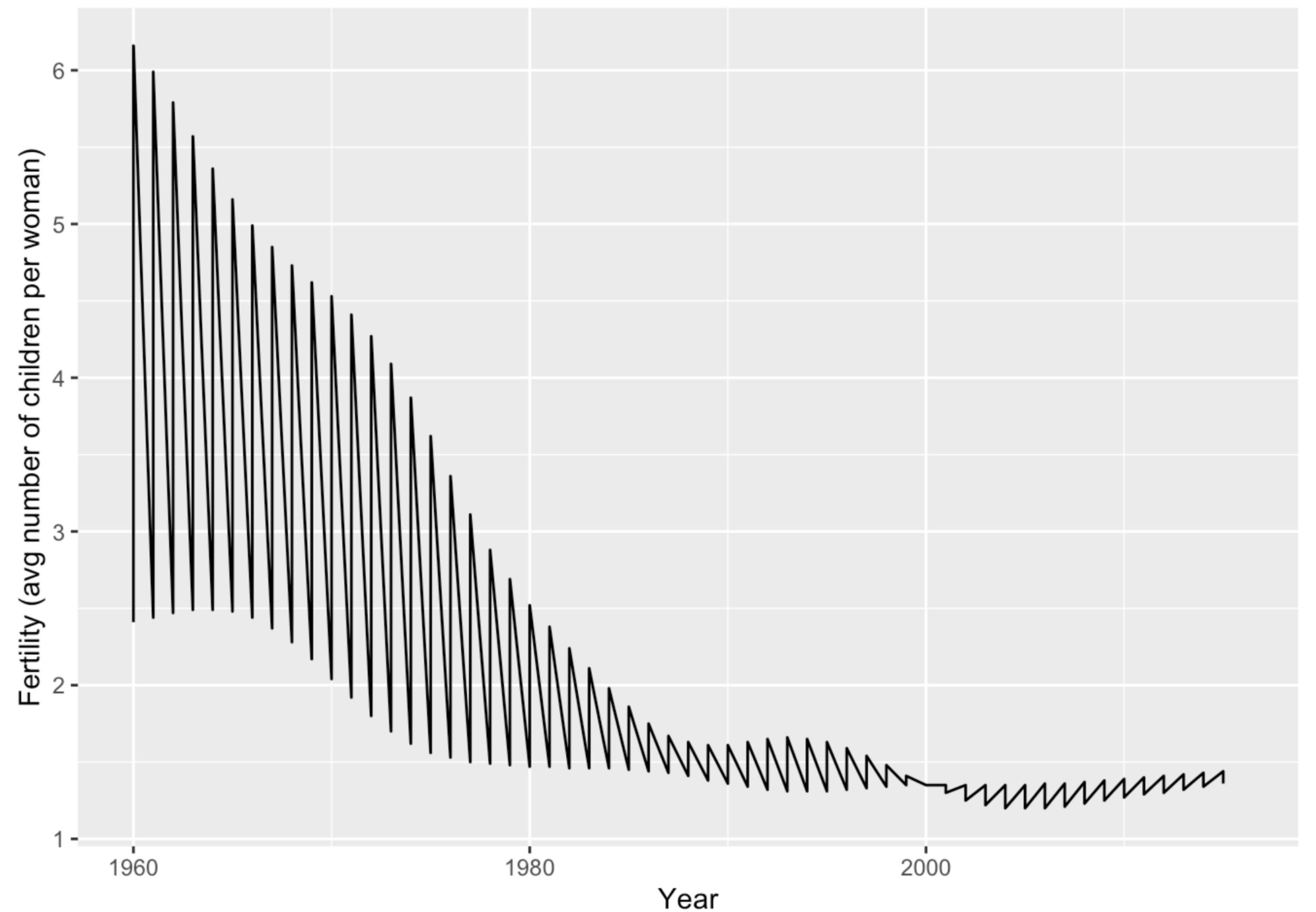


Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```

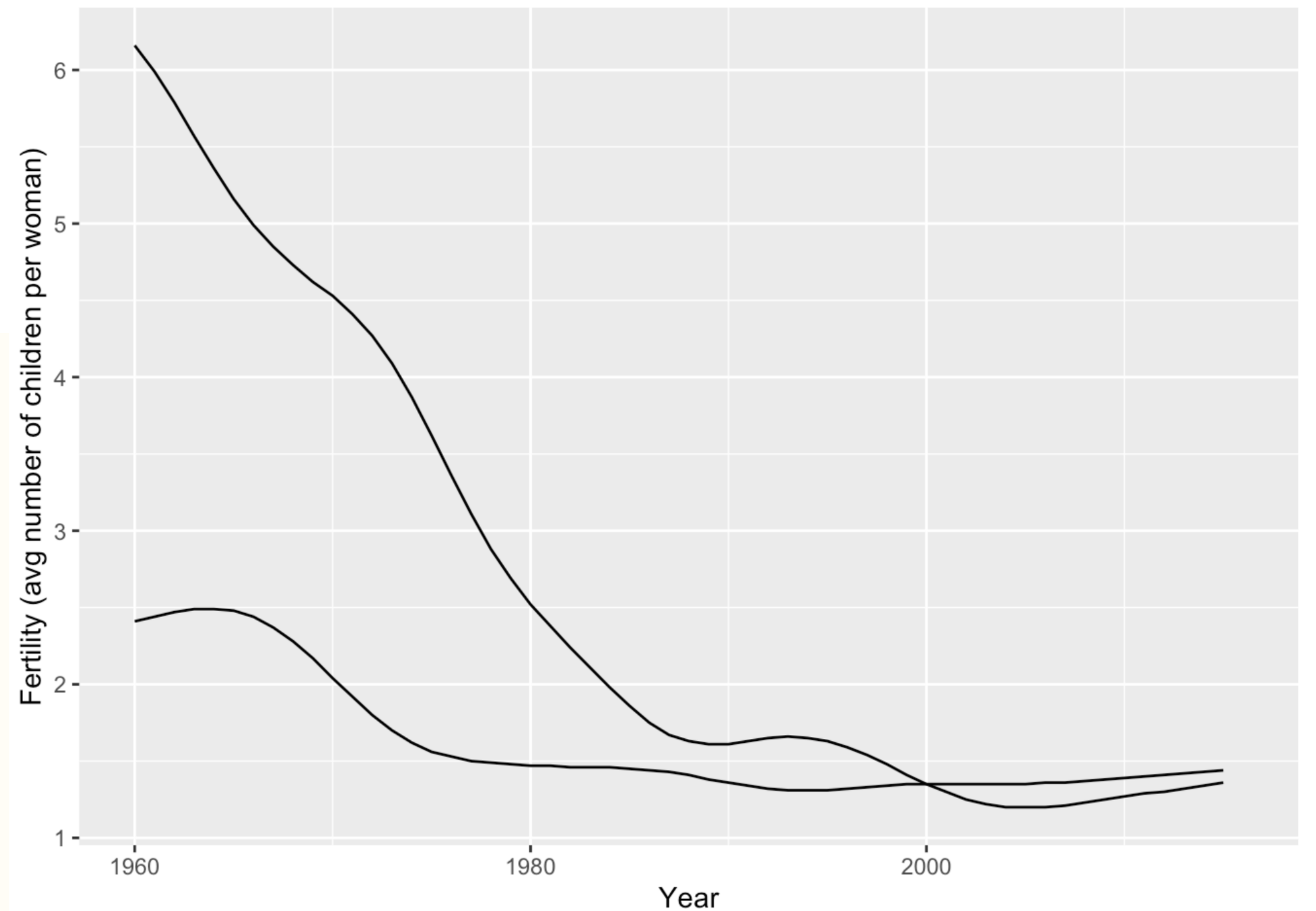
Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility)) +
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```



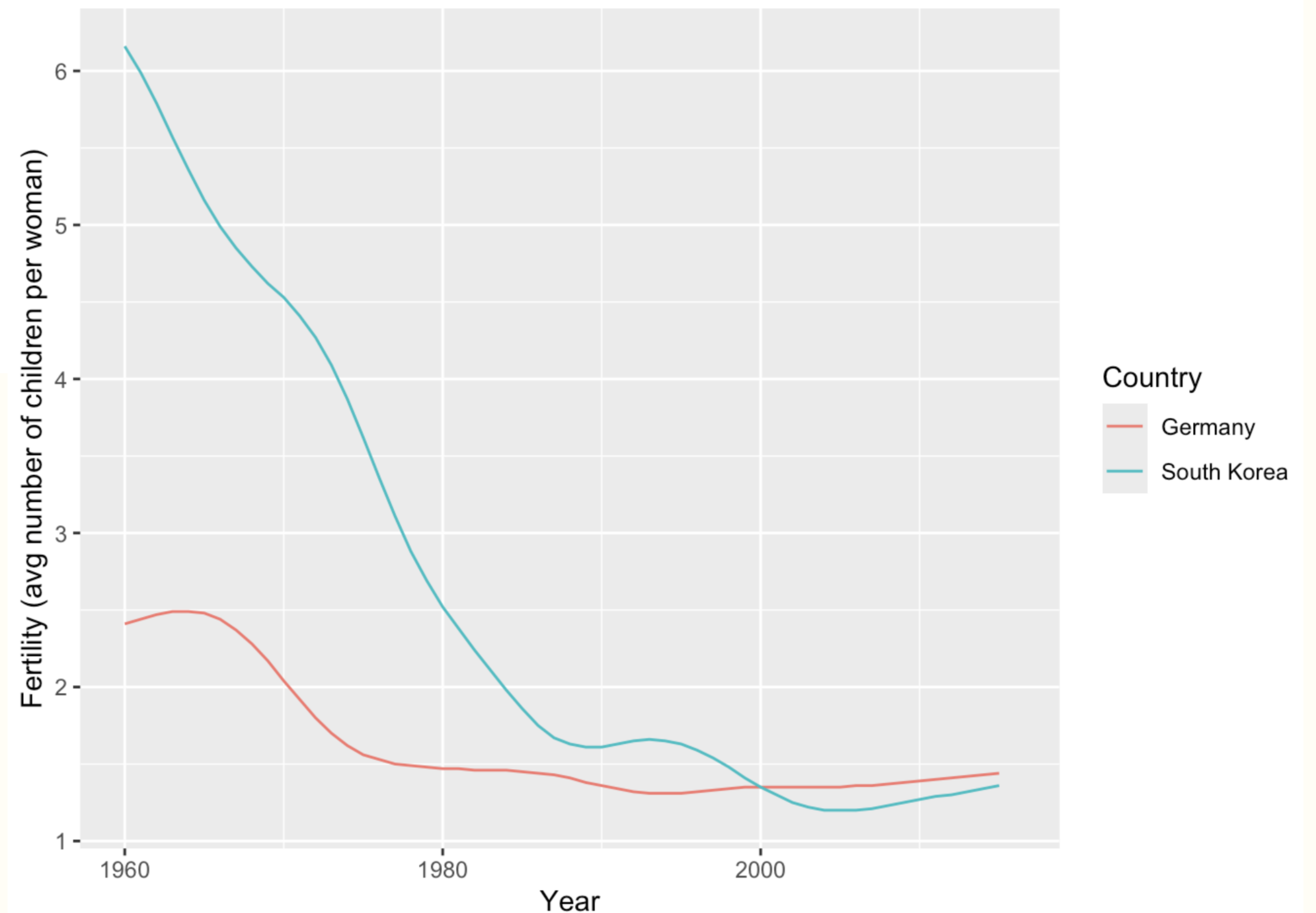
Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility, group = country)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)")
```



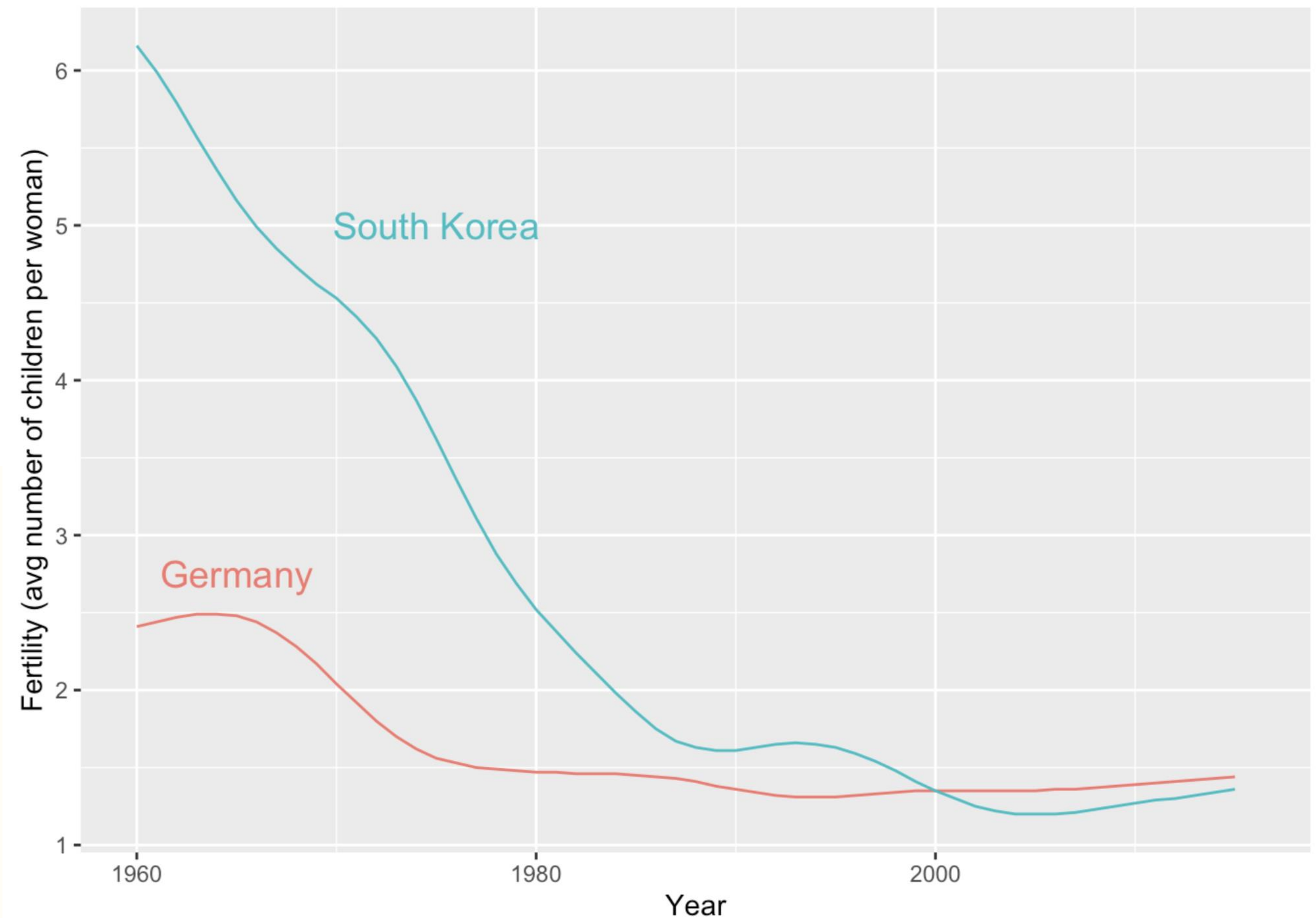
Time series plots

```
countries <- c("South Korea", "Germany")
gapminder %>% filter(country %in% countries) %>%
  ggplot(aes(year, fertility, color = country)) +
  geom_line() +
  xlab("Year") +
  ylab("Fertility (avg number of children per woman)") +
  scale_color_discrete(name = "Country")
```



Labels over legends

```
labels <- data.frame(country = countries,  
                     x = c(1975, 1965),  
                     y = c(5, 2.75))  
  
gapminder %>% filter(country %in% countries) %>%  
  ggplot(aes(year, fertility, color = country)) +  
  geom_line() +  
  geom_text(data = labels, aes(x, y, label = country), size = 5) +  
  theme(legend.position = "none") +  
  xlab("Year") +  
  ylab("Fertility (avg number of children per woman)")
```



Summary

- Time series plots are line plots that show trends over time
- The **group** or **color** arguments must be defined to plot one line per group or category
- Using different colors for different groups or categories eases interpretability
- When only a few lines are plotted, we prefer labels over a legend

Data visualization case study

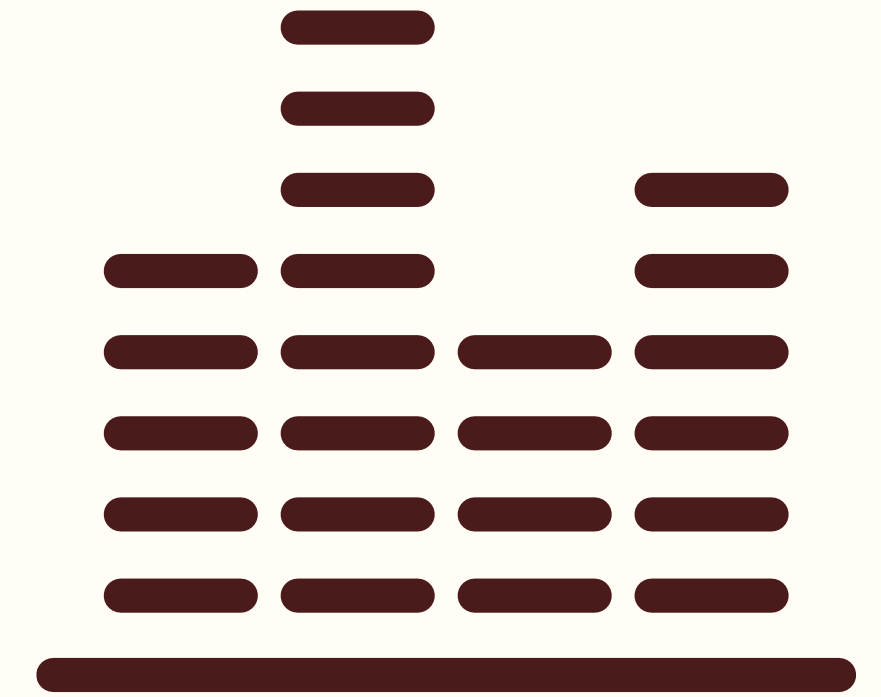
Transformations

Heather Mattie, PhD

Transformations

Data transformations and axis transformations can make it easier to understand patterns, relationships, and distributions, making a visualization **easier to interpret**

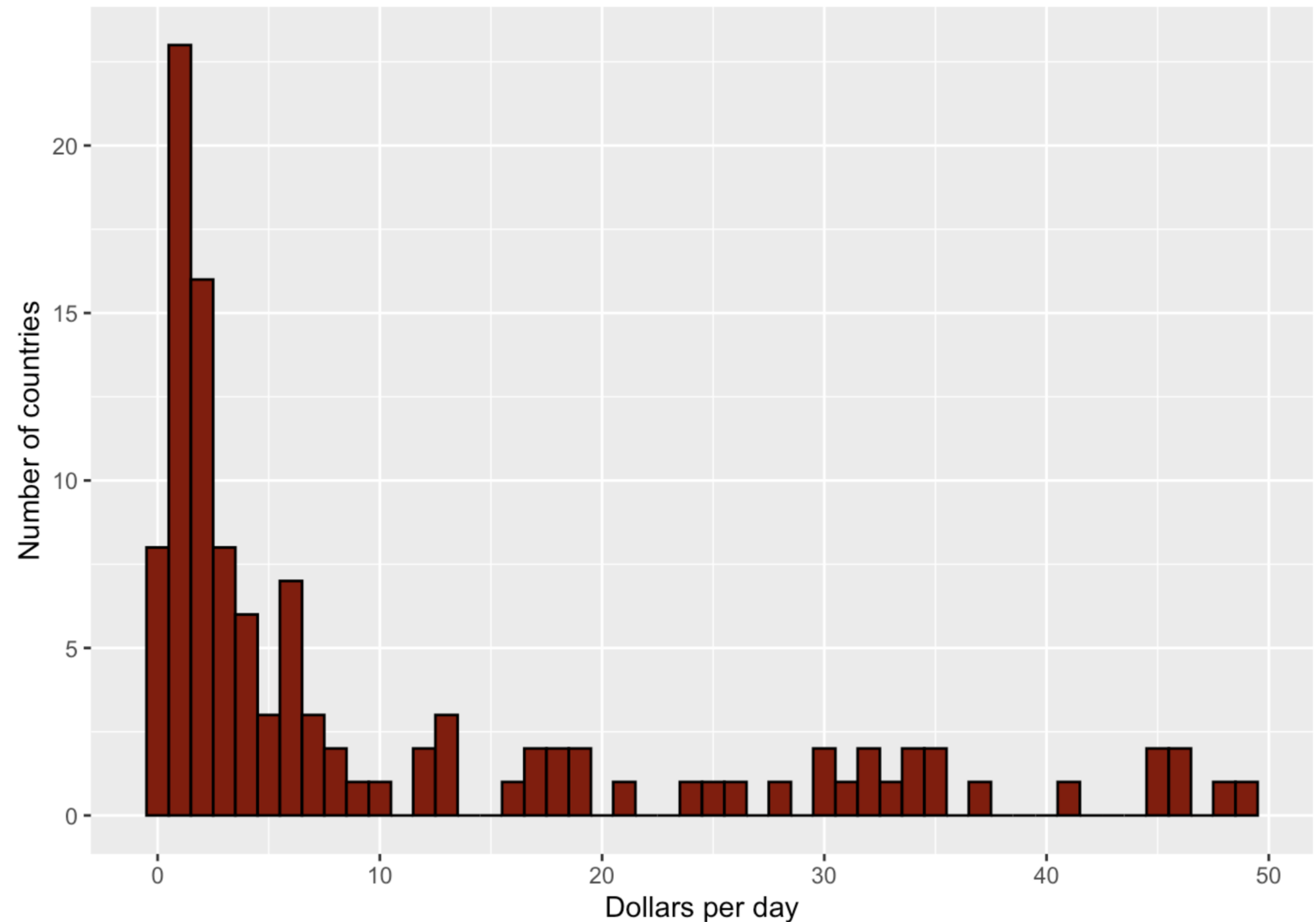
- Data transformations involve mathematical operations on the data
- Axis transformations change the scale and values that appear on a plot axis



Country income distribution

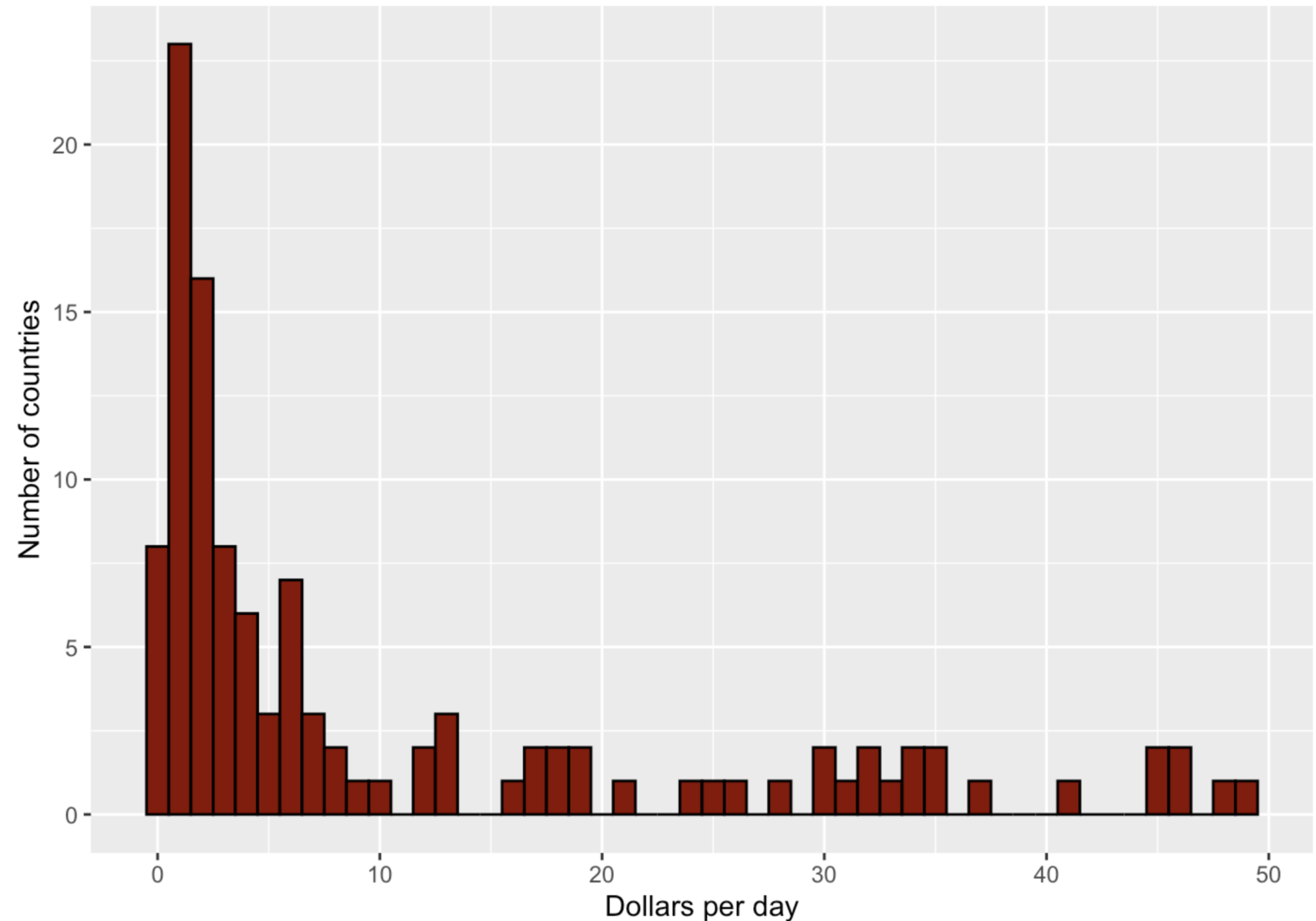
Let's visualize the distribution of country incomes using the **gapminder** data

- The **gdp** column contains the total **gdp** in dollars for a particular country in a particular year
- The values have been adjusted for inflation so comparisons can be made across years



Country income distribution

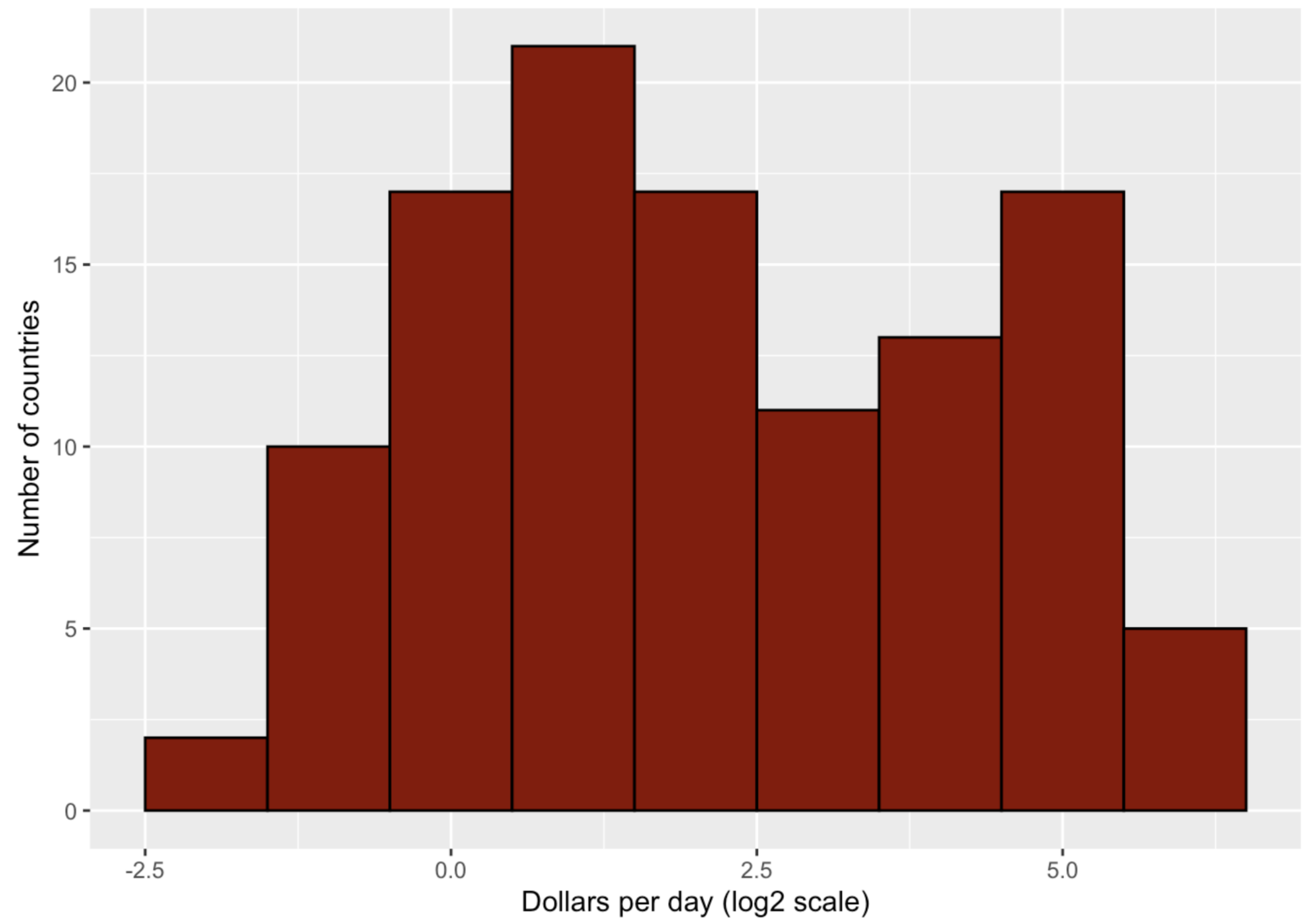
```
gapminder <- gapminder %>%  
  mutate(dollars_per_day = gdp/population/365)  
  
past_year <- 1970  
  
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black", fill = "darkred") +  
  xlab("Dollars per day") +  
  ylab("Number of countries")
```



Data transformation

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(log2(dollars_per_day))) +  
  geom_histogram(binwidth = 1, color = "black", fill = "darkred") +  
  xlab("Dollars per day (log2 scale)") +  
  ylab("Number of countries")
```

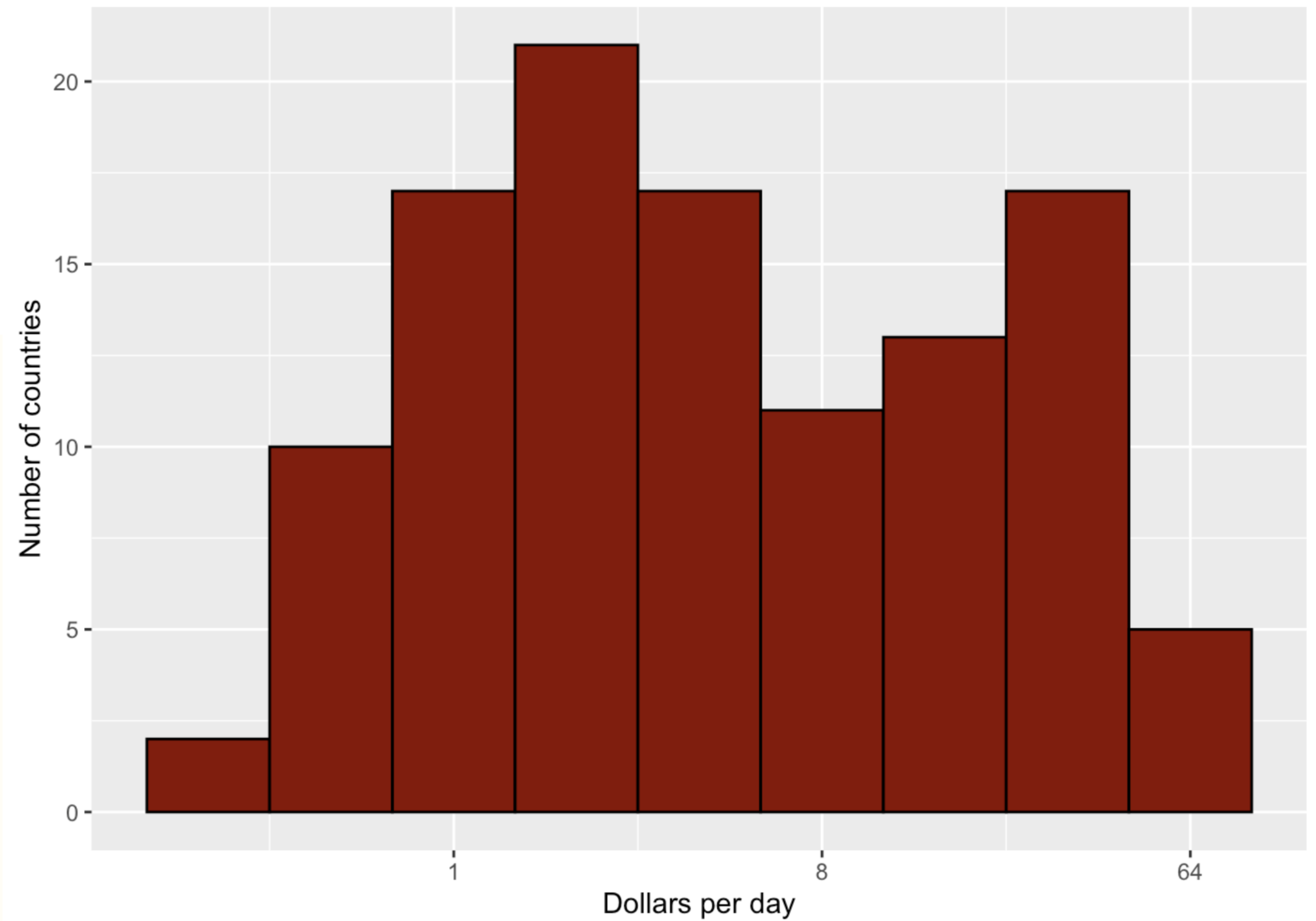
Data transformations change
the values of the data directly



Axis transformation

```
gapminder %>%  
  filter(year == past_year & !is.na(gdp)) %>%  
  ggplot(aes(dollars_per_day)) +  
  geom_histogram(binwidth = 1, color = "black", fill = "darkred") +  
  scale_x_continuous(trans = "log2") +  
  xlab("Dollars per day") +  
  ylab("Number of countries")
```

Transforming an axis changes the values and scale of an axis but plots the original data values, which is usually easier to interpret



Summary

- Data and axis transformations can improve data visualizations and make them easier to interpret
- Data transformations change the values of the data
- Axis transformations change the values and scale of a plot axis